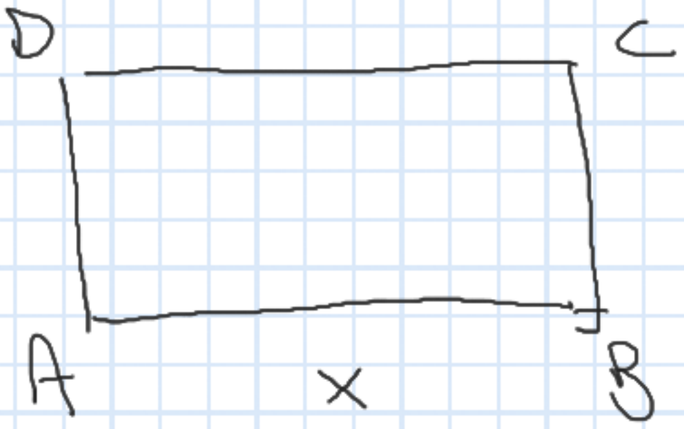
CORREZIONE PROBLEMI

ALGEBRA: pag. 661 n. 552

Figure



$A = \overline{AB} \cdot \overline{AD} = 216\ cm^2$

$2p = 2\overline{AB} + 2\overline{AD} = 60\ cm$

$\overline{AB} = ?$ $\overline{AD} = ?$

$p = \overline{AB} + \overline{AD} = 30\ cm$

$30 = x + \overline{AD}$

Incognite

$\overline{AB} = x$ $\overline{AD} = 30 - x$

C.A.

$x > 0$

Equazione e risoluzione

$$A = \overline{AB} \cdot \overline{AD}$$

$$216 = x \cdot (30 - x)$$

$$x(30 - x) = 216 \quad \rightarrow \quad 30x - x^2 - 216 = 0$$

$$x^2 - 30x + 216 = 0$$

$$\frac{\Delta}{4} = \left(\frac{b}{2}\right)^2 - ac = \left(\frac{-30}{2}\right)^2 - 216 =$$

$$= 225 - 216 = 9$$

$$x_{1,2} = \frac{-\frac{b}{2} \pm \sqrt{\frac{\Delta}{4}}}{a} = \frac{-\left(-\frac{30}{2}\right) \pm \sqrt{9}}{1} = +15 \pm 3 = \begin{cases} 15 - 3 = 12 \\ 15 + 3 = 18 \end{cases}$$

Risposte

Le misure dei due lati del rettangolo sono

$\overline{AB} = 18$ cm e $\overline{AD} = 12$ cm

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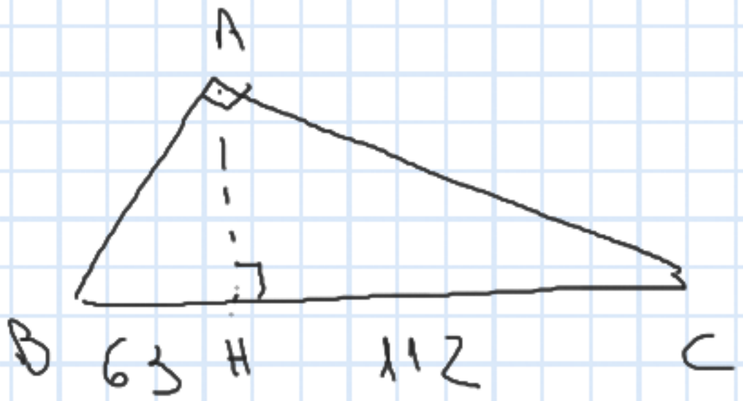

$\overline{BH} = 63\,cm$

$\overline{HC} = 2\,\overline{BH} - 14\,cm.$

$\overline{HC} = 2 \cdot 63 - 14 = 112\,cm$

$A_{(ABH)} = \dfrac{\overline{BH} \cdot \overline{AH}}{2} = ?$

2° Teorema di Euclide:

$\overline{AH}^2 = \overline{BH} \cdot \overline{HC}$

$$\overline{AH} = \sqrt{\overline{BH} \cdot \overline{HC}} = \sqrt{63 \cdot 12} = \sqrt{7056} = 84 \text{ cm.}$$

$$A_{(ABH)} = \frac{\overline{AH} \cdot \overline{BH}}{2} = \frac{\overset{42}{\cancel{84}} \cdot 63}{\cancel{2}} = 2646 \text{ cm}^2$$